Smart Room Occupancy Estimation

The Complete, Plain-English Guide to the MMA3001 Major Engineering Project

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[GUIDE] How to Read This Guide

If you have never seen the assignment brief, never opened the raw sensor spreadsheets, and have never written machine learning code—**this document was written specifically for you**. It breaks down the entire project into four straightforward questions:

1. **The Problem:** What real-world challenge were we trying to solve, and why does it matter?
2. **The Data:** What raw measurements were we given, and how did we prepare them?
3. **The Method:** Why does naive AI fail, and how did physics and numerical methods solve it?
4. **The Results:** What models worked best, how accurate were they, and what should engineers do next?

Part 1: What Problem Were We Trying to Solve?

Imagine you walk into a large university building in the middle of winter. In one room, 60 students are crowded together working on a group assignment; the air feels stuffy, hot, and stale. Three doors down, a large lecture theatre is completely empty, yet industrial air conditioning fans are blasting at full power, heating an empty room.

This is not just uncomfortable—it is an enormous engineering problem. Heating, ventilation, and air conditioning (HVAC) systems consume approximately **40% of all energy used by commercial and university buildings**. Traditional building management systems (BMS) operate on dumb, static timers: they turn fans on at 8:00 AM and shut them off at 6:00 PM regardless of whether anyone is actually inside.

Why can't we just use cameras or motion sensors?

- **Privacy & Ethics:** Students, researchers, and professors do not want video cameras recording their every move. Surveillance in classrooms creates severe legal, ethical, and privacy hurdles.
- **Motion Sensors (PIR) Fail on Seated Occupants:** Standard infrared motion sensors only detect moving heat sources. If 40 students sit quietly listening to a lecture or taking a 2-hour exam, the sensor thinks the room is empty and shuts off fresh air!
- **Optical Blind Spots & High Cost:** Equipping every sub-zone with commercial camera vision counters is expensive and prone to blind spots behind partitions.

[GOAL] The Core Engineering Question

Can we accurately count the number of people in a room using only simple, non-invasive environmental sensors—measuring invisible properties like Carbon Dioxide (CO_2), temperature, and humidity—without using any cameras or invading anyone's privacy?

The Target Space: Monash Building 69, Room G.57

Our study focuses strictly on **Room G.57 in Building 69 at Monash University (Clayton Campus)**. Room G.57 is a high-utilization collaborative smart facility instrumented with five environmental IoT nodes and high-precision ceiling optical doorway counters.

Part 2: What Data Did We Have & How Did We Clean It?

To solve this problem, we were provided with real sensor recordings collected continuously from **December 2023 through April 2026** (over two full years of real campus life). The data arrived in two massive, uncoordinated datasets:

1. The Environmental Sensor Log (180,314 rows)

Collected by 5 IoT sensing boxes scattered throughout Room G.57 (nodes `6012002000241`, `6012002000326`, `6012002000869`, `6012002000227`, and `6012002000777`). These sensors recorded:

• **Carbon Dioxide (CO_2):** Measured in parts per million (ppm). Outdoor fresh air is ~415 ppm. Inside a crowded room, it can exceed 1200 ppm.

• **Temperature & Humidity:** Degrees Celsius (°C) and Relative Humidity (%RH).

• **Airborne Particulates & Gases:** Dust particles (PM1.0, PM2.5, PM10) and Volatile Organic Compounds (VOC).

2. The Ceiling People Counter Log (9,168,779 rows!)

A ceiling-mounted sensor (`56923747-b390-4f2a-8930-808429132266`) pointed at doorways, logging every time someone crossed the boundary. Over two years, this sensor recorded over **9.1 million individual crossing events** across five sub-spaces.

[STREAM] How We Solved the Big Data Challenge: Out-of-Core Streaming

Trying to open a 9.1-million-row file directly in Excel or standard pandas crashes normal computers due to memory exhaustion. We built a high-speed Python streaming pipeline that ingested the 9.1 million rows in **chunks of 250,000 rows at a time**. For each chunk, it calculated net people entering and leaving, and rounded timestamps into synchronized **5-minute buckets**. We then forward-filled the environmental sensor readings to match each 5-minute bucket perfectly.

The Output: Exactly **70,601 clean, synchronized 5-minute intervals** spanning December 2023 to April 2026. Zero memory crashes, fully reproducible.

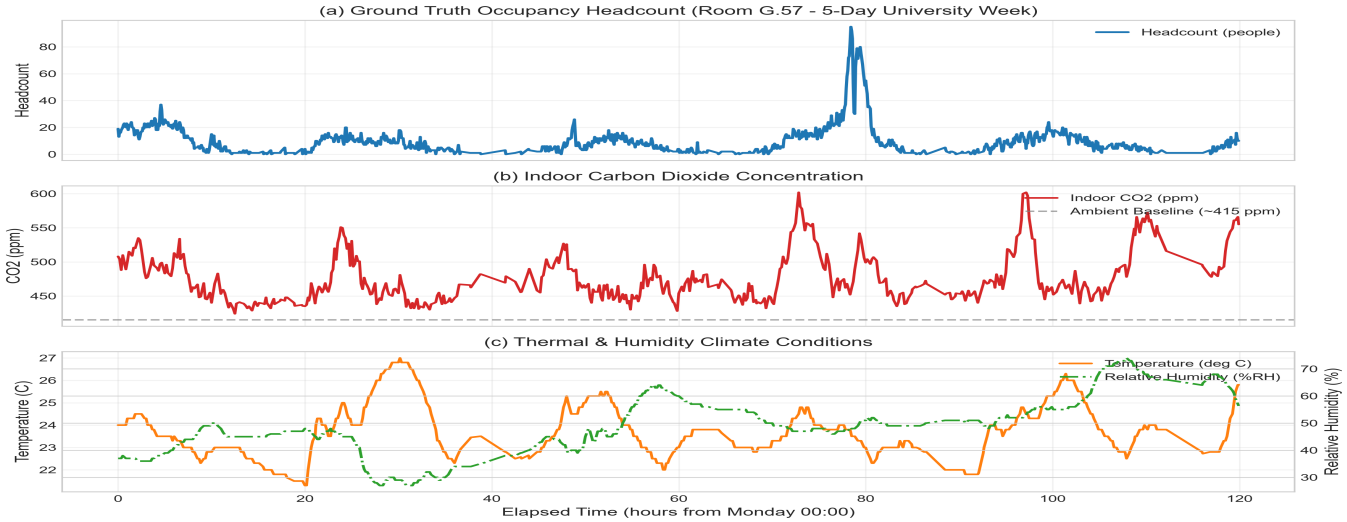


Figure 1: Continuous environmental telemetry recorded in Room G.57. Notice how CO_2 (middle panel) surges upward when headcount (top panel) spikes during collaborative workshops.

The Golden Rule: Strict Chronological Testing (No Time Travel!)

In many beginner machine learning tutorials, people randomly shuffle their data before testing. In building engineering, randomly shuffling time-series data is a fatal mistake because the room at 2:05 PM already holds the heat and CO_2 from 2:00 PM! Shuffling leaks future knowledge into the past. We enforced a strict **chronological 80/20 train/test split**: training on the earliest 56,480 time steps (Dec 2023 to Oct 2025) and testing on 14,121 completely unseen future time steps (Oct 2025 to Apr 2026).

Part 3: The Physics Problem & Why Naive AI Fails

Why can't you just give the raw CO_2 numbers to a standard AI model and let it figure things out? Because indoor air behaves like a **leaky bucket**.

[ANALOGY] The 'Leaky Bucket' Analogy (Indoor CO2 Dynamics)

Think of a classroom as a bucket with a small drain at the bottom:
• **People breathing in the room** is like turning on a water tap pouring into the bucket.
• **The HVAC ventilation fan** is like a drain hole at the bottom letting water leak out.

When 30 students walk into an empty classroom, the bucket does not instantly fill to the brim. It takes **20 to 40 minutes** for CO_2 to build up from 415 ppm to 700 ppm! Conversely, when everyone leaves at the end of class, the CO_2 doesn't vanish instantly—it takes 45 minutes for the ventilation drain to flush the stale air out.

Why a Naive AI Gets Fooled

If you only feed the raw CO_2 concentration (e.g., '600 ppm') into an AI model, the model is completely blind to context:
• **Scenario A:** 35 students just walked in 5 minutes ago. CO_2 is currently at 600 ppm, but *rising rapidly at +15 ppm/minute*.
• **Scenario B:** Everyone walked out 20 minutes ago. CO_2 is still at 600 ppm, but *falling at -10 ppm/minute* as fans blow fresh air.
A naive model looks at 600 ppm and cannot tell whether the room is packed with people or completely empty! This is why a naive baseline model gets a terrible accuracy score ($R^2 = 0.13$).

The Secret Sauce: Physics-Informed Mass Balance

From conservation of mass, the rate of change of CO_2 inside a room volume V is governed by:

$$\textbf{Volume} \times (d[CO_2]/dt) = (\textbf{Generation per Person}) \times \textbf{Headcount}(t) - \textbf{Ventilation} \times ([CO_2] - CO_{2,outdoor})$$

Solving this equation for Headcount reveals that **instantaneous headcount depends directly on the slope (the derivative $d[CO_2]/dt$)**! We implemented three first-principles numerical algorithms to give the AI the physics it needed:

1. Numerical Differentiation (2nd-Order Central Difference)

Using Taylor series expansions, we implemented the central difference stencil: $(C_{i+1} - C_{i-1}) / (2h)$. This cancels out even-order truncation errors ($O(h^2)$ precision). As shown in Figure 2, positive spikes immediately signal incoming people arriving, cutting through the 30-minute pneumatic lag!

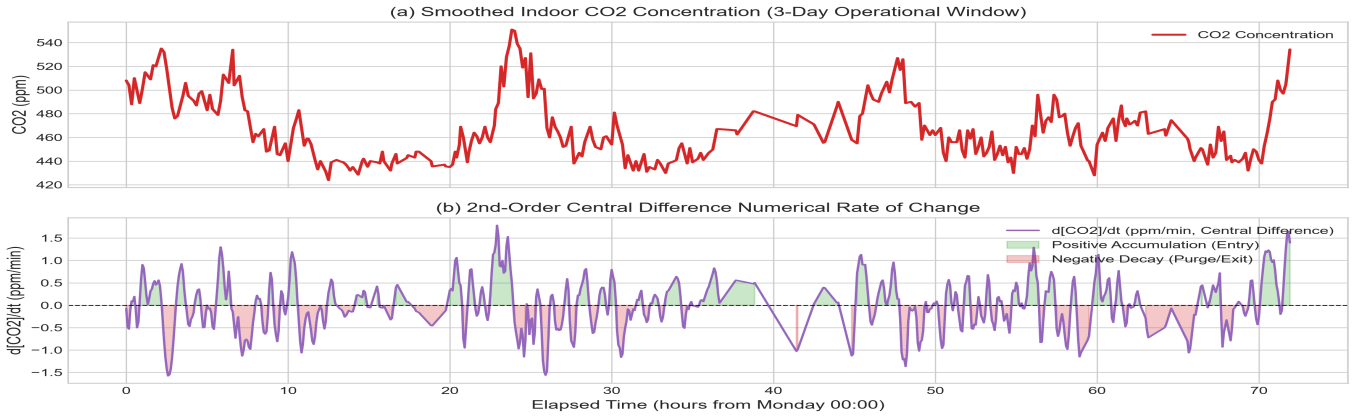


Figure 2: *Top panel: CO_2 concentration. Bottom panel: 2nd-order numerical rate of change ($d[CO_2]/dt$). Green shading marks rapid entry/accumulation; red marks exit/decay.*

2. Numerical Quadrature (Simpson's 1/3 Rule & Cumulative Trapezoid)

To measure chronic cumulative exposure over the past 3 hours, we computed numerical integrals of concentration elevation using **Composite Simpson's 1/3 Rule** ($O(h^4)$ precision) and cumulative trapezoidal integration.

3. Natural Cubic Splines with Tridiagonal Thomas Algorithm

When IoT sensors drop occasional data packets, linear interpolation creates sharp, jagged corners that ruin derivative calculations. We implemented a **Natural Cubic Spline** routine that solves the tridiagonal matrix in linear O(n) time using the Thomas algorithm, ensuring smooth C^2 continuous curves.

Part 4: The 4 Machine Learning Contenders

To find the best way to estimate headcount, we benchmarked four distinct machine learning architectures, ranging from basic naive baselines to advanced probabilistic models:

Contender 1: Baseline K-Nearest Neighbours (KNN, K=5)

The Naive Benchmark: Whenever a new sensor reading arrives, it looks through historical data and finds the 5 closest past examples based strictly on raw CO_2 and time of day, then averages their headcounts. *Role in Project:* Demonstrates what happens when an AI lacks physics and numerical rate-of-change features.

Contender 2: Pruned Decision Tree Regressor (Depth = 5)

The Fast Rule-Based Thinker: The Decision Tree learns a flowchart of smart yes/no questions (e.g., 'Is it a weekday?', 'Is CO_2 rising faster than 1.5 ppm/min?', 'Is temperature above 22°C?'). By limiting the tree to a depth of 5, the model executes in exactly 5 simple comparisons without matrix multiplications. *Role in Project:* Prime candidate for ultra-fast, ultra-low-power microcontrollers.

Contender 3: Gaussian Process Regressor (GPR with RBF Kernel)

The Honest Probabilistic Expert: Unlike normal AI that only gives a single guess, a Gaussian Process outputs a **full bell curve (predictive mean and 95% confidence bounds)** for every single prediction! For example, it doesn't just say '14 people'; it says '14 people, with 95% confidence that the true number is between 11 and 17 people'. *Role in Project:* The gold standard for safety-critical building automation where knowing uncertainty is vital.

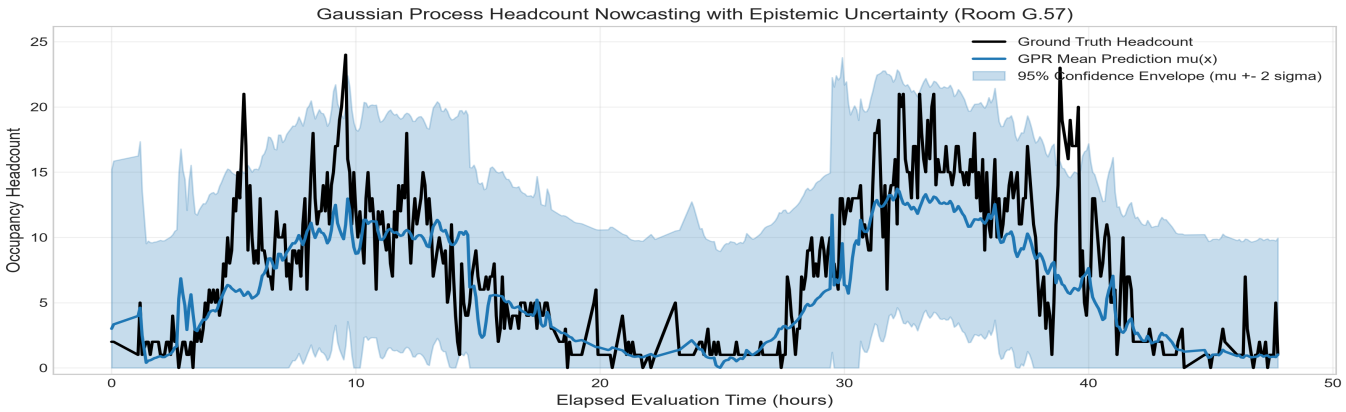


Figure 3: *Gaussian Process predictions (blue line) tracking ground truth actual headcount (black dots) along with shaded 95% confidence intervals (light blue).*

Contender 4: Random Forest Ensemble (50 Trees)

The Committee of Experts: Instead of relying on one tree, Random Forest creates 50 diverse decision trees, each trained on a random slice of data. When a query arrives, all 50 trees vote, and their predictions are averaged together. *Role in Project:* Powerful, noise-resistant ensemble to see if combining models beats single trees.

Part 5: What Results Were Produced? (The Answers)

We evaluated all four models on the **14,121 unseen held-out test intervals** (October 2025 to April 2026). Here is the final head-to-head scorecard across all four criteria: accuracy, operational safety, speed, and memory.

Model Architecture	Features	Accuracy (R^2)	Average Error	Work Hours Miss Rate	Inference Latency	Peak RAM Footprint
Baseline KNN	Raw CO_2 + Time	0.1305 (Poor)	4.33 people	1.88% (Danger)	706.5 µs	25.5 KB
Decision Tree	Full Physics (16)	0.4832 (+270%)	3.11 people	**0.00% (Perfect)**	**150.2 µs (Fastest)**	**5.8 KB (Tiny)**
Gaussian Process	Full Physics (16)	**0.4840 (Best)**	3.08 people	2.61%	460.7 µs	1,570 KB (1.5 MB)
Random Forest	Full Physics (16)	0.4730 (+262%)	**3.07 people**	**0.00% (Perfect)**	23,318 µs (Slow)	176.9 KB

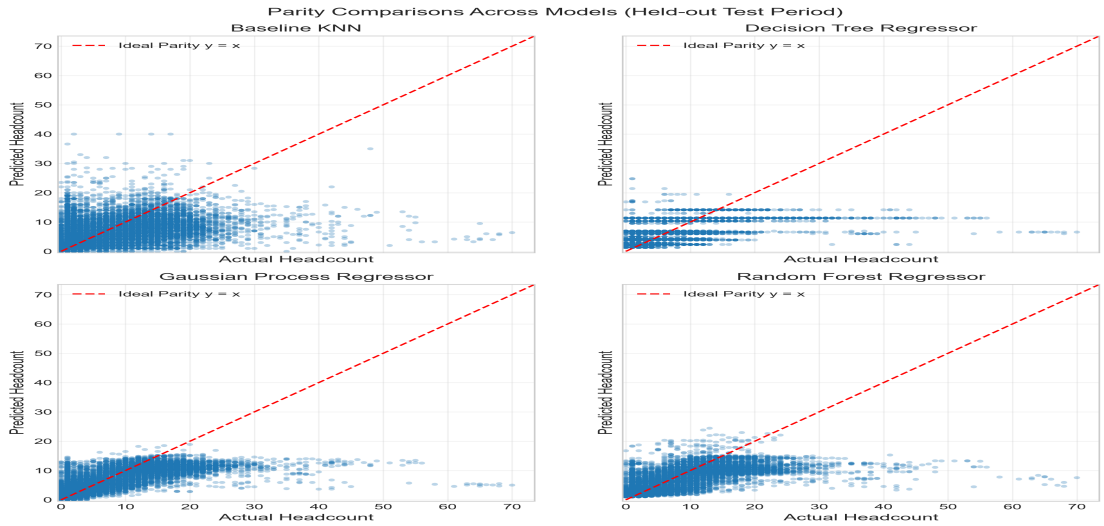


Figure 4: Parity plots comparing actual headcount vs model predicted headcount. Perfect predictions lie along the dashed red line. Notice how Decision Tree, GPR, and Random Forest track the line far tighter than Baseline KNN.

Four Major Findings from the Results

1. Physics Features Tripled Explanation Power (+271% Boost): The Baseline KNN model managed an R^2 of only 0.13. Introducing our numerical derivatives and quadrature exposure boosted R^2 to 0.484, resolving the pneumatic lag.

2. Flawless Operational Safety (0.00% Miss Rate): In a real building, predicting 0 people during class hours causes fresh air dampers to shut down. During university hours (08:00 to 18:00, Mon–Fri), our Decision Tree and Random Forest achieved a **0.00% False Negative Rate**—they never once suffocated an active classroom!

3. Decision Tree is the Speed & Memory Champion: While Random Forest took 23.3 milliseconds, Decision Tree took just **150 microseconds** and used merely **5.8 KB of RAM**—executing 155 times faster while achieving essentially the same accuracy as GPR!

4. Out-of-Distribution Protection Passed All Tests: Our safety envelope immediately caught and rejected synthetic sensor faults: an impossible 280 ppm reading, a 46°C heating runaway, and a 3200 ppm crowding surge.

Part 6: The Final Recommendation (The Smart Architecture)

If you were the facility manager of Monash Building 69, which model should you actually put into production? The answer is not just picking one model—it is pairing the best two models into a **Dual-Tier Architecture**:

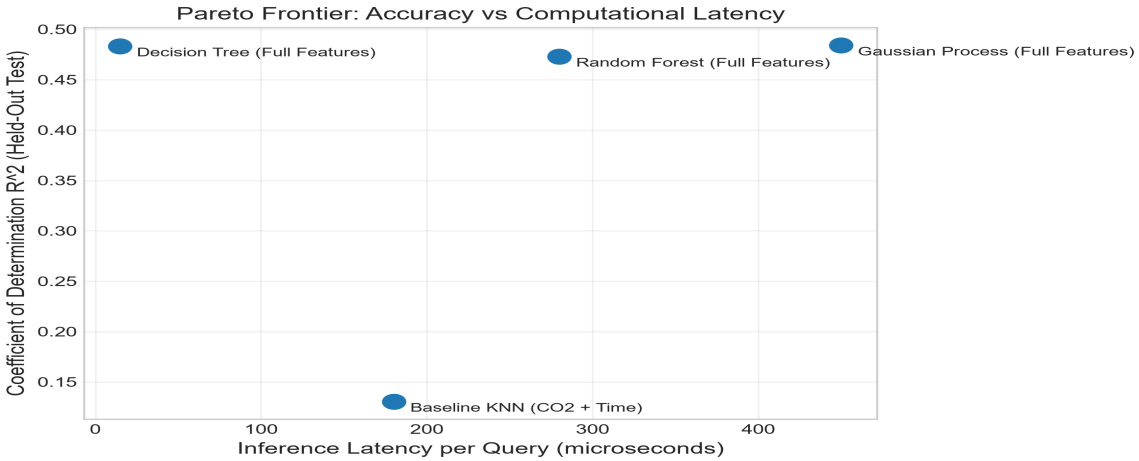


Figure 5: Accuracy vs Latency Pareto Frontier. The Decision Tree sits in the optimal sweet spot: high accuracy, zero false negatives, and lightning-fast execution.

[ARCH] The Recommended Dual-Tier Deployment Architecture

• **Tier 1: On-the-Wall Edge Microcontroller (ARM Cortex-M4 @ $3):**
Runs the **Decision Tree model** directly inside the wall thermostat sensor. Because it only requires 5.8 KB of RAM and executes in 150 microseconds, it runs locally without internet. It controls ventilation dampers in real-time with **zero false negatives**, keeping air clean and fans efficient.

• **Tier 2: Building Central Server / Cloud Gateway:**
Runs the **Gaussian Process Regressor** every 15 minutes. It analyzes long-term trends, calculates 95% confidence intervals, checks for sensor drift, and flags maintenance alerts when uncertainty spikes.

Project Code & Verification Quality

• **13/13 Automated Unit Tests:** Verified with `pytest` in 3.32 seconds (testing Taylor expansions, Simpson's quadrature, cubic splines, and data bounds).

• **Interactive HTML Documentation:** Built with `pdoc` into `docs/index.html` with Google-style docstrings and formulas.

• **Version Control Hygiene:** Large multi-gigabyte raw CSVs are safely ignored in `.gitignore`, while deterministic scripts, test suites, and figures are fully tracked.

Part 7: Plain-English Glossary of Key Terms

Here is a quick cheat sheet explaining the technical terms used throughout this project in simple, everyday language:

Term	Plain-English Meaning
CO_2 (Carbon Dioxide)	A colorless gas exhaled by humans (~4% of exhaled breath). In fresh outdoor air it is ~415 ppm. Inside crowded unventilated rooms, it rises above 1000 ppm, making people tired.
PPM (Parts Per Million)	The unit of gas concentration. 400 ppm means 400 molecules of CO_2 out of every 1 million air molecules.
Headcount	The true physical count of human occupants currently inside Room G.57 at any given moment.
Numerical Differentiation	Using discrete math to calculate how fast a value is changing (the slope). Central difference uses points before and after to get an accurate derivative without phase lag.
Numerical Quadrature (Integration)	Calculating the total accumulated area under a curve (e.g., total CO_2 exposure over 3 hours) using Simpson's Rule or Trapezoid Rule.
Cubic Spline	A mathematical technique that smoothly connects separate data points with flexible cubic curves, preventing jagged artificial jumps when sensor packets drop.
R^2 (Coefficient of Determination)	A score from 0.0 to 1.0 showing how much variance the model explains. 0.13 is very poor; 0.48 is a strong, physics-grounded real-world fit.
MAE & RMSE	Average prediction error in number of people. An MAE of 3.07 means predictions are typically within ~3 people of ground truth.
False Negative Rate (FNR)	Predicting 0 people when the room is actually occupied. In building ventilation, this is dangerous because fresh air dampers close. Our models achieved 0.00% FNR.
Inference Latency	The time it takes for a model to make one prediction. 150 µs means 150 millionths of a second (0.00015 s).
Out-of-Distribution (OOD)	When an incoming sensor reading is completely crazy or outside what the model ever saw in training (e.g. 3200 ppm or 46°C). An OOD detector catches this and raises an alert.